

the training set. On closer inspection, we find that the most similar environments to the majority of the example configurations are clusters of water, ammonia and NO_2 molecules. Several examples contain oxygen-bearing molecules near water molecules, allowing for sampling of hydrogen bonding. We also find configurations containing clusters of large aromatic compounds containing carbon, nitrogen, oxygen and hydrogen. Interestingly, we also find a large cluster containing diverse configurations of small C,H,N,O-containing fragments, including carboxyl and amide fragments, *e.g.* mp-997182.

Performance summary

Stable MD at ambient conditions, but incorrect free energy of conformers.

A.31 Molecule-Surface Interactions

A.31.1 Adsorption energies - S24 dataset

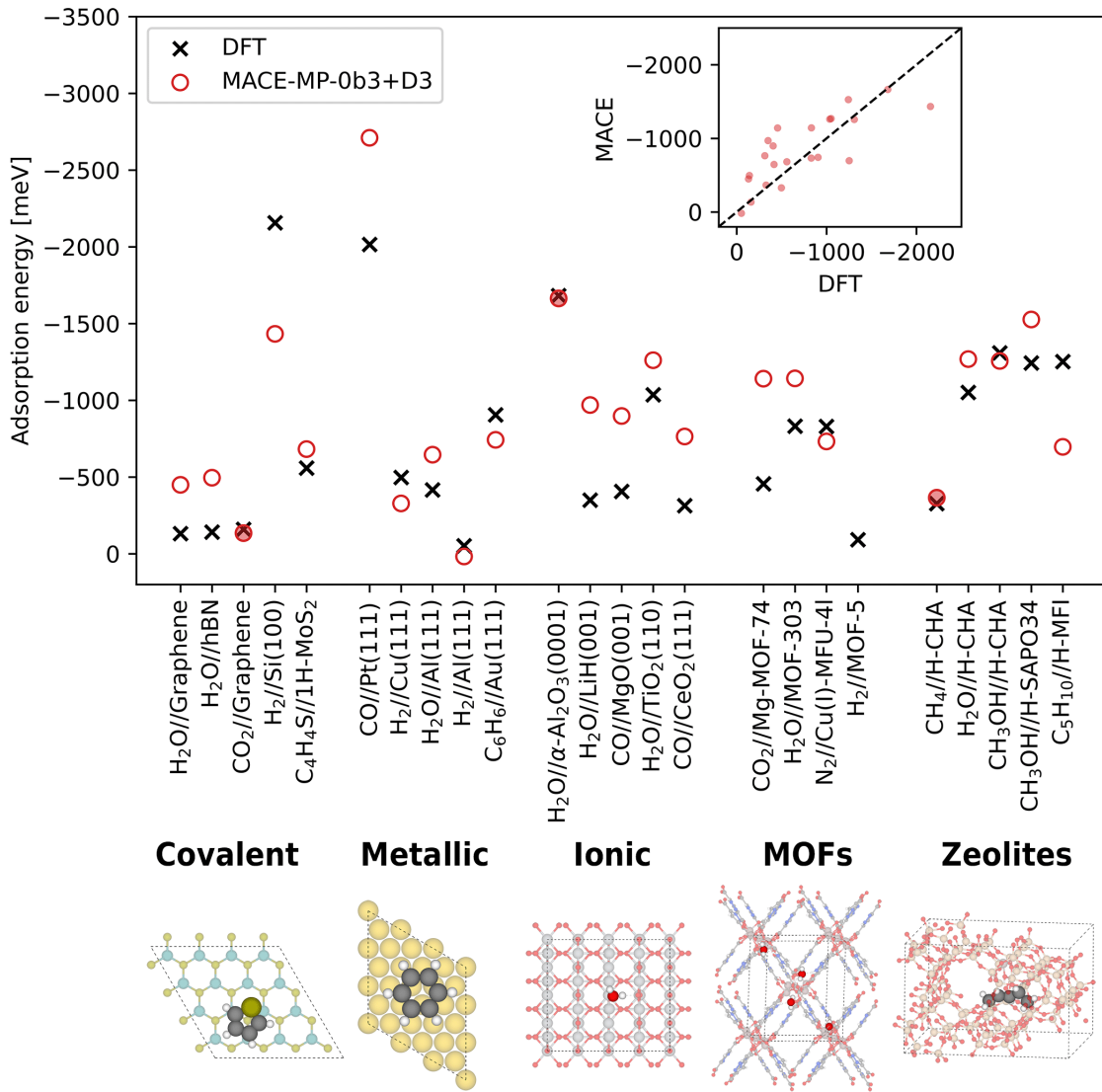


Figure S48: Comparison between DFT (PBE+D3(BJ), red crosses) and MACE-MP-0b3+D3 (blue circles) adsorption energies calculated for a diverse set of surfaces. These surfaces consist of covalent, metallic, and ionic bonds as well as the porous material classes: MOFs and zeolites. Filled circles indicate surfaces where the MACE model reaches chemical accuracy (43 meV) agreement to DFT. The inset shows that there is a good correlation between MACE-MP-0b3+D3 and the DFT references.

Describing the interaction between a molecule and a surface with first-principles accuracy is central towards designing new and improved materials for heterogeneous catalysis, gas storage and separation, and many more (283). Here, we test the accuracy of MACE-MP-0 augmented with the Becke-Johnson D3 dispersion correction for a set of prototypical systems found in surface chemistry, encompassing metallic, covalent and ionic-bonded surfaces, together with porous metal-organic frameworks (MOFs) and zeolites. The structures used within this work - dubbed the S24 dataset - were taken from an amalgamation of published (284–294) and unpublished works. Fig. S48 summarises the computed PBE-D3(BJ) adsorption energies using PYMATGEN to generate VASP (4, 295, 296) inputs with the MPRelaxSet settings.

Table S4: Mean absolute deviation in meV for the S24 dataset and its sub-categories of surfaces.

	MACE-MP-0b3+D3
Covalent	308
Ionic	361
MOF	380
Metal	265
Overall	305
Zeolite	229

We summarise the overall MAD as well as the MAD within the sub-categories of surfaces for MACE-MP-0b3+D3 in Table S4. Performance is moderately good, with an overall MAD of 305 meV, in line with the errors observed from a typical DFT functional. Performance across the various sub-categories of surfaces is relatively uniform, with best performance for zeolites with an MAD of 229 meV. It performs poorly for H₂ on Si(100), with errors exceeding 600 meV.

Similarity statement

The S24 dataset contains 72 structures comprising 24 adsorbates, surfaces, and adsorbate-surface combinations. Based on the UMAP analysis, the elemental compositions of the adsorbates occur 6 to 27 times, the surfaces 0 to 9043 times and the adsorbate-surface combinations 0 to 4896 times on their own or along with other elements. Importantly, the training dataset does not contain any gas-phase molecules, surface-truncated models, or MOFs. For bare surfaces, on average, similar element compositions occur in the training data as follows: 3997 covalent, 5805 metallic, 2859 ionic, 46 MOFs, and 88 zeolites. For adsorbate-surface combinations, on average, the number of similar element compositions occurring in the training data are lowered to: 1492 covalent, 340 metallic, 499 ionic, 46 MOFs, and 59 zeolites. There is no clear correlation between the number of similar training data for a surface type and the accuracy of MACE-MP-0b3+D3.

Performance summary

Moderately good agreement is generally achieved for adsorption energies, with improved performance for adsorption onto zeolites for MACE-MP-0b3+D3. Performance appears to become poorer with dissociative chemisorption of H₂ on Si(100), albeit still reproducing the adsorption energy to the correct order of magnitude.

A.31.2 Relative energies - OC157 dataset

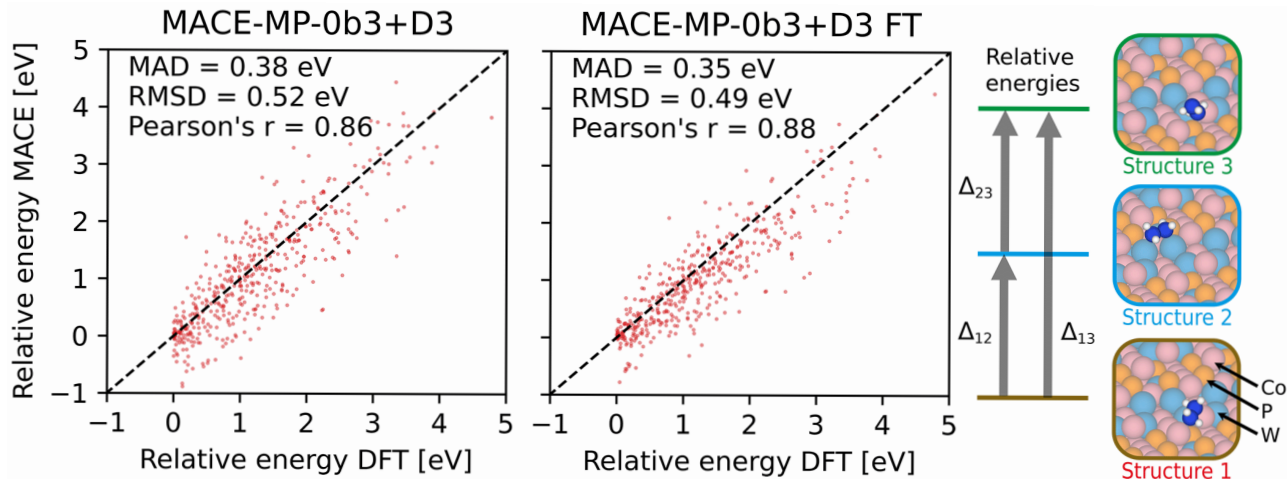


Figure S49: Comparison of MACE-MP-0b3+D3 and its fine-tuned (FT) version (see text) against DFT (PBE-D3(BJ)) in predicting the relative energies between three structures across 157 molecule-surface combinations. This contains a highly diverse set of molecule-surface combinations which span 54 elements of the periodic table involving up to three elements per surface.

Identifying the most stable structures for a molecule-surface system is pivotal towards predicting the activity and selectivity of a catalyst, facilitating rational design of new catalyst materials (297). We compare MACE-MP-0b3+D3 and the model following fine-tuning against `MPRelaxSet` DFT at predicting the relative energies between 3 structures for 157 molecule-surface combinations. The fine-tuned (FT) models were trained on a dataset containing one high-energy structure (not from the test set) for each of the 157 molecule-surface combination. These surfaces were taken from the Open Catalyst Challenge 2023 (298, 299), using structures generated by the baseline EquiformerV2 model trained on the OC20-S2EF-2M dataset. While 200 molecule-surface combinations were originally provided, we have excluded those containing oxygen (O) in combination with several transition metals (Co, Cr, Fe, Mn, Mo, Ni, V and W) as this leads to complications with the Hubbard U correction (see main text) with `MPRelaxSet` settings. Fig. S49 shows the 471 relative energies (Δ_{12} , Δ_{13} and Δ_{23}) for each of the remaining 157 molecule-surface systems, where Δ_{XY} is the relative energy between structure X and Y, predicted with the MACE models and calculated with DFT. Overall, we observe a strong correlation between MACE-MP-0b3+D3 and DFT on the relative energies, providing a Pearson correlation coefficient of 0.86 and an MAD of 0.38 eV. We find that fine-tuning produces a small improvement, decreasing the MADs by 0.03 meV. In particular, out of the 157 molecule-surface combinations, the lowest DFT energy configuration was correctly identified by MACE-MP-0b3+D3 for 126 of the surfaces. Fine-tuning further increases this number to 132 for MACE-MP-0b3+D3 FT.

Similarity statement

The molecule-surface combinations in the OC157 dataset cover all elements up to and including Bi (atomic number 83), except He, Li, Be, B, F, Ne, Mg, Cl, Ar, Br, Kr, I, Xe, Ba and all lanthanoids. The UMAP analysis shows that 126 of the 157 molecule-surface combinations have no similarity to the training dataset, with a further 20 having less than 10 and only 3 having more than 20 similar compositions.

Performance summary

Good performance at predicting the absolute value of the relative energies between various molecule-surface configurations. Lowest energy (DFT) structure correctly predicted in 80% of the cases for MACE-MP-0b3+D3 with fine-tuning further improving performance to 84%

A.32 Computational efficiency (twenty-element alloy)

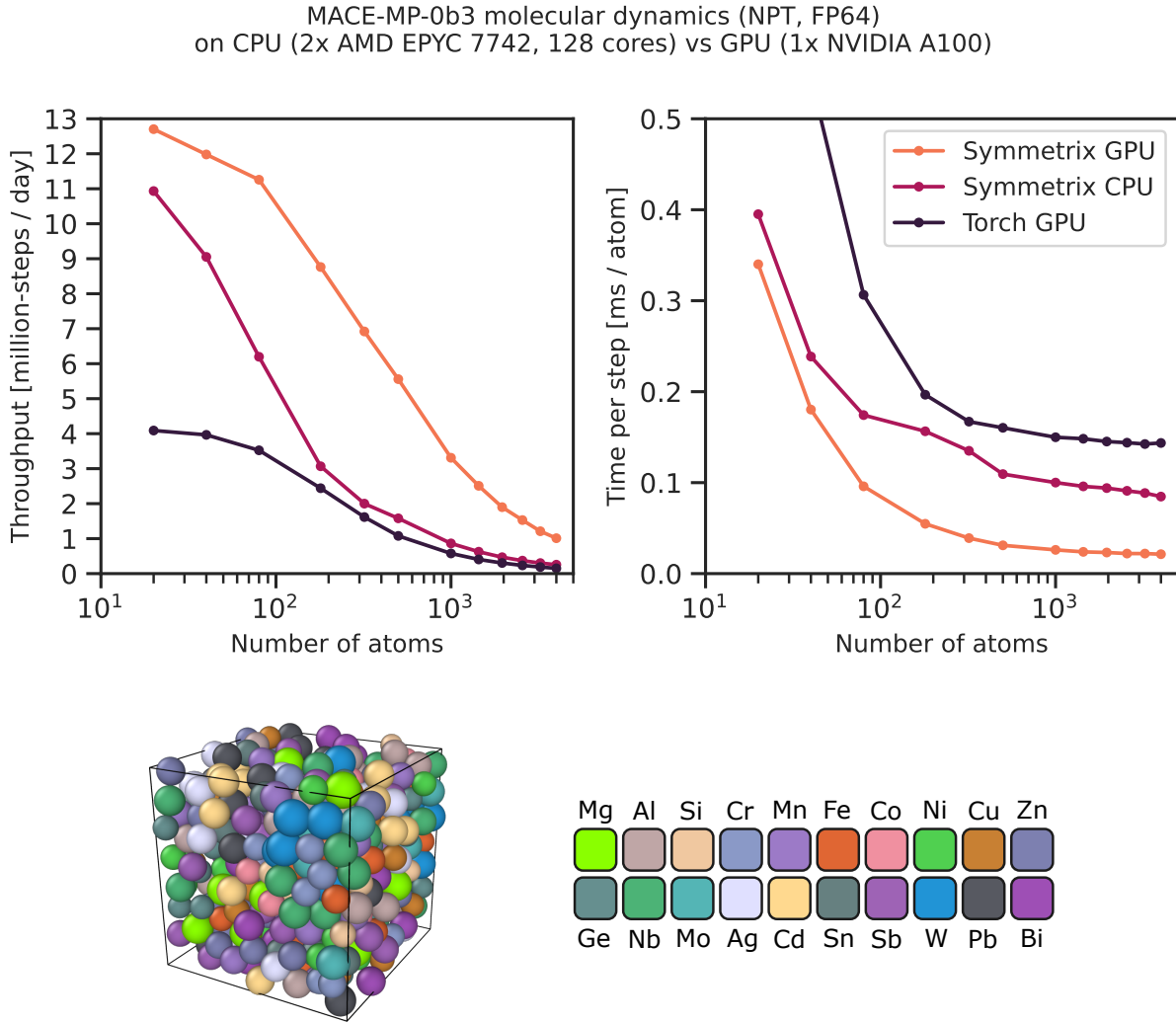


Figure S50: Double-precision computational performance for NPT dynamics in LAMMPS, measured for the 20-element Cantor alloy at ambient pressure and 300 K. The two plots show the same data, highlighting throughput (left) and speed (right). Results are shown for a 128-core CPU node utilizing AMD EPYC 7742 CPUs as well as a single NVIDIA A100 GPU. Each plot has three curves: one for the Torch baseline on GPU and two for an optimized MACE evaluator available from the `symmetrix` package. The `symmetrix` evaluator uses Kokkos, a vendor-agnostic library, to enable fast CPU and GPU performance.

This section demonstrates the computational performance of the medium MACE-MP-0b3 model (without dispersion correction), while also illustrating stable dynamics for a large, diverse system. High entropy alloys are multicomponent mixtures with at least four or five distinct elements, where each component appears in non-trace proportions. Here, we consider the 20-component alloy investigated by Cantor (300), which contains equal amounts of Mn, Cr, Fe, Co, Ni, Cu, Ag, W, Mo, Nb, Al, Cd, Sn, Pb, Bi, Zn, Ge, Si, Sb, and Mg. [See also recent computational work by Ceriotti and co-workers (109, 301).] Our simulations were performed with LAMMPS (302), utilizing its Kokkos extensions (303), on AMD CPUs and NVIDIA A100 GPUs.

Figure S50 shows single-node computational performance and Figure S51 demonstrates multi-node scaling. For each simulation, we began with a 20-atom FCC primitive cell and generated randomised supercells